be smoothed with 320-400 grit emery cloth but do not try to salvage the shaft if nicks or wear is severe.

INPUT GEAR AND PLANETARY CARRIER

Check the teeth on the gear. Minor nicks can be dressed off with an oilstone but replace the gear if any teeth are broken, cracked, or chipped. The bearing surface on the gear can be smoothed with 300-400 grit emery cloth if necessary.

Examine the carrier body and pinion gears for wear or damage. The carrier will have to be replaced as an assembly if the body, pinion pins, or pinion gears are damaged. Inspect the plastic washer on the inside, bottom of the planetary carrier. The washer is trapped by the pinion gears and the complete planetary will need to be replaced if the washer is worn or damaged.

SHIFT FORKS / HUBS / SLEEVES

Check condition of the shift forks and mode fork shift rail. Minor nicks on the shift rail can be smoothed with 320-400 grit emery cloth.

Inspect the shift fork wear pads. The mode and range fork pads are serviceable and can be replaced if necessary.

Check both of the sleeves for wear or damage, especially on the interior teeth. Replace the sleeves if wear or damage is evident.

DRIVE CHAIN

Examine the drive chain and shaft bearings. replace the chain if stretched, distorted, or if any of the links bind. Replace the bearings if rough, or noisy.

LOW RANGE ANNULUS GEAR

Inspect annulus gear condition carefully. The gear can be serviced separately.

FRONT - REAR CASES

Inspect the cases for wear and damage. Replace the input retainer seal, do not reuse it.

Check case condition. If leaks were a problem, look for gouges and severe scoring of case sealing surfaces. Also make sure the front case mounting studs are in good condition.

Check the front case mounting studs and vent tube. The tube can be secured with Loctite™ 271 or 680 if loose. The stud threads can be cleaned up with a die if necessary. Also check condition of the fill/drain plug threads in the rear case. The threads can be repaired with a thread chaser or tap if necessary. Or the threads can be repaired with Helicoil™ stainless steel inserts if required.

OIL PUMP / OIL PICKUP

Examine the oil pump pickup parts. Replace the pump if any evidence of debris or damage exists. Do not disassemble the pump from the rear case half as individual parts are not available and it will not be possible to reassemble the pump cover to the rear case half. The pump is only available as part of the rear case half assembly. The pickup screen, hose, and tube are the only serviceable parts and are available separately.

ASSEMBLY

BEARINGS AND SEALS

(1) Position the input gear bearing in the input retainer bearing bore. Spread the snap-ring captured in the retainer with suitable snap-ring pliers and press the bearing into place. Verify that the snap-ring has engaged the groove in the bearing.

(2) Start front output shaft bearing in case (Fig. 46). Then seat bearing with Handle C-4171 and Installer 6953.

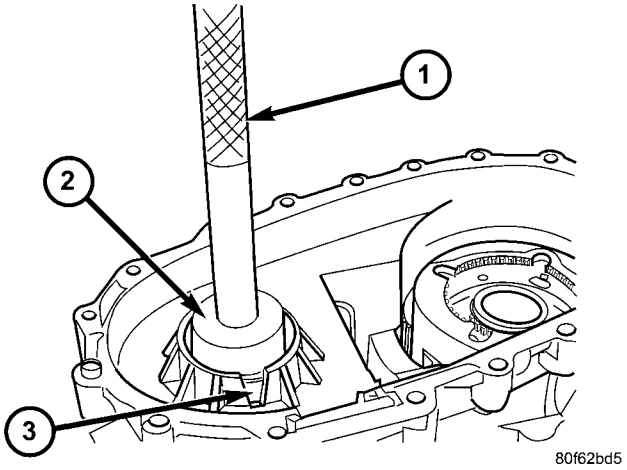

Fig. 46 Front Output Shaft Bearing Installation

1 - HANDLE C-4171
2 - INSTALLER 6953
3 - BEARING

(3) Install the front output shaft bearing retaining ring.

(4) Install the rear output shaft bearing into the rear case half with Installer 6953 and Handle C-4171 (Fig. 47).